Sai Krishna Bugga

bt24btech11009@iith.ac.in | bskrishnabugga5@gmail.com | +91 8985186921 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.52
Sri Chaitanya Junior Kalasala	Class XII (Telangana Board of Intermediate Education, Hyderabad)	2024	97.90%
SREE CHAITANYA HIGH SCHOOL	Class X (Board of Secondary Education)	2022	10.00%

SKILLS

Languages : C , R , Python

Frameworks : Geopandas, numpy, matplotlib, pyvmd, PySDkit, MDanalysis, Bio.seq, ggplot2, DESeq2.

Technologies : Visual Molecular Dynamics (VMD) , PyMOL , Solid Edge , graphPad Prism

Transcriptomics: DESeq2, featureCounts, Pathway Analysis

- **Tools**: FastQC - BWA - SAMtools - BCFtools - fastp - seqtk - featureCounts

RESEARCH EXPERIENCE

Research Internship - IIT Bombay (Department of Computer Science & Engineering)

- Worked on ***Data Gathering and Annotation Systems for ML Models***, contributing to dataset collection and annotation for computer vision applications. Gained hands-on experience with ML data pipelines, annotation tools, and AI solutions for real-world surveillance and drone-based monitoring.

- Awarded a **Letter of Recommendation** by Prof. Ganesh Ramakrishnan in recognition of outstanding performance, technical contributions, and research potential during the internship.

PRACTICAL EXPERIENCE

Basic Biotechnology Laboratory | Practical Experience

- Cultured *E. coli* DH5 using streak plating and secondary inoculation; monitored bacterial growth and plotted growth curves using OD measurements.
- Performed DNA **PCR amplification**, **gel electrophoresis**, and **SDS-PAGE**, followed by gel visualization using a **Gel Documentation System (Gel Doc)**.
- Developed hands-on skills in aseptic techniques, microbial culture, molecular biology, and experimental data interpretation.

PROJECTS

Next Generation Sequencing (NGS) Data Analysis

- Analyzed high-throughput sequencing data using standard NGS workflows.
- Performed quality control, sequence alignment, variant calling, and data interpretation.
- Used bioinformatics tools such as FastQC, BWA/Bowtie2, SAMtools, GATK, and IGV.
- Interpreted sequencing results and generated reports for genomic variant analysis.

P vs NP, Integer Factorization, and Encryption of Genomic Data

- Analyzed the computational complexity of integer factorization and its role in RSA-based encryption for securing genomic data.
- Developed Python programs using SymPy to benchmark factorization of composite integers and evaluate runtime growth across increasing input sizes.
- Investigated the impact of quantum algorithms (Shor's algorithm) and post-quantum cryptography on the future security of genomic information.

Comparative Transcriptomic Analysis of Normal and Cancer Samples

- Performed comparative analysis of normal and cancer transcriptomic datasets to identify differentially expressed genes and associated molecular patterns.
- Processed and analyzed RNA-seq data using **FastQC**, **HISAT2**, and **DESeq2** in a Linux environment.
- Conducted quality control, sequence alignment, differential gene expression analysis, and interpreted results through graphical visualizations and biological insights, Presented findings using graphical summaries and biological interpretation.

Tools: FastQC, HISAT2, DESeq2, BWA (if you actually used it)

Operating System: Linux

RELEVANT COURSES

Introduction to Programming , R and Python for Biologists , Data Structures & Applications , Biostatistics ,Basic Bioinformatics ,Biological Databases and Big Data Biology ,Next Generation Sequencing (NGS)

POSITIONS OF RESPONSIBILITY

Head of Events | Prayas X RDC

- Led the end-to-end planning, coordination, and execution of multiple club events by managing event timelines, cross-domain collaboration, resource allocation, and event documentation.
- Successfully organized and managed a flagship event with an audience of approximately **600 attendees**, demonstrating strong leadership, organizational, and event management skills.

Hospitality Coordinator - Inter IIT Sports Meet 2025

EXTRACURRICULAR

Volunteer in National Service Scheme (NSS)

Cycling